

Kalhara Tennakoon

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PROFILE

First Class Computer Science graduate, top of the Software Engineering batch at ESU Kandy, with industry experience in cloud-native systems and DevOps at IFS R&D International. My undergraduate work applies machine learning to real problems: a veterinary clinic system with ML forecasting (Grade A) whose local RAG assistant reached the Ascentic AI Launch Pad Top 10, and a study predicting course difficulty from student evaluations. I am seeking graduate study in machine learning and reliable, secure systems infrastructure.

Research Interests: MLOps & Machine Learning Systems · LLM Applications & Retrieval-Augmented Generation · AI Infrastructure · Agentic AI Systems · Cloud Security & DevSecOps

EDUCATION

Kingston University

BSc (Hons) Computer Science (Software Engineering) — First Class Honours
Delivered via ESU Kandy, Sri Lanka

London, United Kingdom

2026

- **Class Rank:** Batch Top – Software Engineering, ESU Kandy
- **Final Year Project:** VetCare Pro – Smart Veterinary Clinic Management System (Grade A, 76%)

RESEARCH & PROJECTS

VetCare Pro – Smart Veterinary Clinic Management System

Oct 2025 – Aug 2026

React, Node.js, Express, PostgreSQL, pgvector, Flask, Scikit-learn, Ollama, SwiftUI

Final Year Project · AI Launch Pad

- Full-stack veterinary clinic management platform, developed as my final year project (Oct 2025 – Mar 2026, Grade A) and then extended with AI features for the Ascentic AI Launch Pad through Aug 2026.
- Final year project: appointments, medical records, billing, and inventory management with role-based access, a RESTful Node.js/Express API, and JWT authentication.
- AI Launch Pad extension: local RAG assistant using locally hosted LLMs and pgvector, routing between semantic retrieval and exact SQL queries, with human-in-the-loop confirmation for database writes.
- AI Launch Pad extension: pet owner portal and SwiftUI companion iOS app; selected for the Top 10.

Predicting Course Difficulty from Student Evaluation Responses

Feb 2025 – Jul 2025

Python, Scikit-learn, Pandas, Matplotlib, Seaborn, SPSS, Machine Learning

Research

- Supervised machine learning models predicting perceived course difficulty from student evaluation datasets using statistical analysis techniques.
- Data preprocessing, exploratory data analysis, and feature engineering on student evaluation datasets.
- Classification model evaluation with cross-validation and statistical significance testing.
- Identified instructor-related attributes as significant predictors through statistical validation and model interpretation.

SolarCast – Personal Solar Energy Forecasting App

Apr 2026 – Present

Python, FastAPI, React, Prophet, Pandas, REST APIs

Ongoing

- Full-stack ML app predicting monthly solar energy generation, grid export, and cash payouts from residential inverter data using time-series forecasting.
- Time-series forecasting with Meta's Prophet, producing 90% confidence interval predictions.
- Chronological train/test ML pipeline to prevent data leakage across forecast windows.
- Interactive React dashboard with CSV upload support and energy analytics visualisations.

Kubernetes Cluster & Container Image Security Scanner

Jul 2020 – Dec 2020

Kubernetes, Docker, Golang, ArgoCD, Rancher, Azure DevOps

Industry · IFS

- Containerized security scanning solution for Kubernetes environments, detecting vulnerabilities in container images and cluster configurations.
- Integrated automated security scanning workflows into CI/CD pipelines to strengthen secure deployment practices.

EXPERIENCE

DevOps Engineering Intern

Jul 2024 – Oct 2024

IFS R&D International (Pvt) Ltd

Colombo, Sri Lanka

- Automated Kubernetes-native CI/CD workflows using Tekton, streamlining deployment processes for critical services.
- Integrated Behave automated testing into Tekton CI/CD pipelines, reducing manual debugging effort by ~30% and improving deployment reliability.
- Optimized pipeline configurations and resolved recurring build issues to improve deployment stability and engineering efficiency.
- Conducted a knowledge-sharing session on testing best practices to support team collaboration and code quality improvements.

Software Engineering Intern	Jul 2020 – Dec 2020
<i>IFS R&D International (Pvt) Ltd</i>	<i>Colombo, Sri Lanka</i>
- Developed a cloud security scanning application using Docker and Kubernetes to improve container platform security and vulnerability assessment workflows. - Integrated security scanning tools into CI/CD pipelines, improving vulnerability detection across deployment stages. - Diagnosed deployment and runtime issues while authoring technical documentation to support system reliability and onboarding processes.	

HONOURS & AWARDS

Top 10 Finalist – Ascentic AI Launch Pad	2026
Selected for the Top 10 and invited to present VetCare Pro at Demo Day; recognised for its local RAG assistant, ML forecasting models and companion iOS app, all built for a real working clinic.	
Batch Top – Software Engineering, ESU Kandy	2026
Ranked first in the Software Engineering cohort of the BSc (Hons) Computer Science programme.	
Two-Time Dan Kohn Scholarship Recipient – KubeCon + CloudNativeCon Europe	2022 & 2025
Awarded the highly competitive scholarship recognising contributions to the cloud-native ecosystem.	
First Runner-Up – Hack:Bit Hackathon	2019
Secured 2nd place in a competition organised in collaboration with Microsoft and Sarvodaya Fusion.	
Top 15 Finalist – HaXmas Hackathon	2018
Selected among 75+ competing teams in a national-level hackathon.	
Finalist – Cisco Packet Riders	2018
Reached the national final of a networking and troubleshooting competition open to students across Sri Lanka.	
Cisco NetRiders	2017
Ranked 4th in Sri Lanka and 118th in the Asia Pacific & Japan region.	

TECHNICAL WRITING & OUTREACH

Medium	Dec 2021 – Present
<i>Technical Writer · kalharatennakoon.medium.com</i>	
- Authored 50+ technical articles on DevOps, cloud-native technologies, and software engineering, reaching 150K+ views. Selected articles: “Building, Dockerizing, and Deploying a CRUD API in Go on Kubernetes,” Level Up Coding, Apr 2025; “Getting Started with Tekton Pipelines: A Beginner’s Guide to CI/CD on Kubernetes,” Level Up Coding, Aug 2024; “Getting Started with CronJobs in Kubernetes,” FAUN.dev, Dec 2021.	

STEMUp Educational Foundation	Aug 2018 – Dec 2020
<i>Volunteer</i>	
Contributed to initiatives promoting STEM education and digital literacy among students and youth communities in Sri Lanka.	

CERTIFICATIONS

Cloud, DevOps & Platform Engineering	
Containers & Kubernetes Essentials — IBM	Feb 2025
GitHub Foundations — GitHub	Jan 2025
Introduction to Kubernetes — The Linux Foundation	Dec 2024
Microsoft Certified: Azure Fundamentals (AZ-900) — Microsoft	Jan 2023
Artificial Intelligence & Data	
Google AI Professional Certificate — Google (Coursera)	Feb 2026
Microsoft Certified: Azure Data Fundamentals (DP-900) — Microsoft	Nov 2024
Microsoft Certified: Azure AI Fundamentals (AI-900) — Microsoft	Apr 2024

TECHNICAL SKILLS

Programming: Python, Java, Golang, JavaScript, C, Bash/Shell
Data & AI: Scikit-learn, Pandas, Prophet, pgvector, Ollama, SPSS, Matplotlib, Seaborn, Jupyter Notebook, Google Colab, Power BI, Data Cleaning, EDA
Cloud & DevOps: Docker, Kubernetes, Tekton, ArgoCD, GitHub Actions, Azure DevOps, CI/CD, Infrastructure as Code (IaC)
Cloud Platforms: Microsoft Azure, AWS, Google Cloud Platform, AKS, GKE, Rancher
Web & Databases: React, Node.js, Flask, FastAPI, REST APIs, PostgreSQL, MySQL, MongoDB
Networking: TCP/IP, DNS, Load Balancing, Firewalls, VPN, OSI Model
Tools & Practices: Git, GitHub, Agile, Scrum, Jira, Technical Documentation